

Meenatchi sundaram N

ROBOT SOFTWARE DEVELOPER

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PROFESSIONAL SUMMARY

Results-driven Robotics Software Developer specializing in ROS2 (Jazzy & Humble), autonomous mobile robot (AMR) development, and simulation-based testing with Gazebo and RViz. Experienced in full-stack robot development spanning URDF modeling, SLAM, Nav2 path planning, and costmap tuning. Passionate about building intelligent robotic systems that bridge simulation and real-world deployment. Active contributor to the ROS Tamil Nadu open-source community.

INTERNSHIP

ROS2 -(Robotics Software), Intern 02/2026 – 05/2026 | Chennai
Worked on ROS2-based AMR development including Nav2 stack integration and costmap tuning
Implemented SLAM pipelines and sensor fusion for autonomous navigation
Assisted in Gazebo simulation setup for validation before hardware deployment

Centre Of Robotics And Automation, Intern 02/2025 – 03/2025 | Anna university
Completed an internship in Robotics And Automation, with hands-on Anna university(MIT) experience in PLC programming and Solidworks

EDUCATION

BACHELOR'S DEGREE-(Robotics And Automation), 2022 – 2026 | Coimbatore
DHAANISH AHMED INSTITUTE OF TECHNOLOGY
CGPA - 7.5%

SKILLS

TECHNICAL SKILL

- **Robotics Frameworks:** ROS 2 (Jazzy, Humble), Nav2, SLAM Toolbox.
- **Programming:** C++ , Python, Bash Scripting.
- **Tools & Visualization:** RViz2, Foxglove Studio, Gazebo.
- **Embedded Systems:** Raspberry Pi, ESP32, Linux (Ubuntu 22/24.04), Micro-ROS.
- **Sensors & Hardware:** LiDAR (YDLidar), IMU, Motor Drivers, Encoders.
- **Communication Protocols:** Topics, Services, Actions, Rosbridge (WebSockets).
- **Full-Stack Robotics:** React.js (Robot UI integration), Git/GitHub.

SOFT SKILL

- **Community Leadership:** Managing a ROS 2 developer group and facilitating technical knowledge exchange.
- **Problem-Solving:** Strong analytical skills in debugging complex hardware-software integration issues.
- **Adaptability:** Quick learner of evolving technologies (e.g., transitioning to ROS 2 Jazzy and Ubuntu 24.04).
- **Technical Mentorship:** Guiding peers in troubleshooting SLAM, Nav2, and system-level bottlenecks.
- **Collaboration:** Experienced in working with cross-functional teams using Git for version control.

CERTIFICATIONS

- Udemy - Self Driving And ROS2 Learn by Doing Plan And Navigation
- DE Drones - Drone Technology
- IIT Madras - Next Gen Robotics IoT AI and Global Control
- NPTEL - Wheeled Mobile Robotics (75%- Elite)

PROJECTS

Automated Metal Detected With Real Time Alerts

Built an automated metal detector using LC sensors with Wi-Fi based real-time alerts for remote security monitoring.

Rocker-Bogie Surveillance Robot for Hazardous Environments

Developed a Bluetooth-controlled Rocker-Bogie AMR featuring SN04-N inductive sensors for hazardous object detection and real-time mobile alerts for remote surveillance.

Autonomous Vacuum Cleaning Robot, (AMR Simulation)

Multi-Mode Navigation: Engineered Custom-area, Waypoint, and Freehand path modes via a React-based mission control UI.

Autonomous Charging: Implemented battery-triggered auto-docking logic and self-charging functionality.

Full-Stack Robotics: Integrated React and ROS 2 using Rosbridge for real-time mapping and trajectory execution.